

Project Submission Details

Project Name:	SafeHire: Verified Credential & Skill-Match Recruitment Portal
Group Members:	1. Muhammad Umar Afzaal (Roll No: 23F-3106) 2. Musa Rehan (Roll No: 23F-3093)
Module Allocation:	Module 1 (Umar Afzaal): Post Verified Opportunity Module 2 (Musa Rehan): Browse & Search Catalog
Live / Local URL:	https://safehire-secdev.vercel.app/ Local: http://localhost:3000/ (Vite)
GitHub Repo:	https://github.com/muhammadiumarafzaal/safehire-secdev

Part 1 — Select Two Complementary Modules

Module 1:	Post Verified Internship Opportunity (Enterprise & Placement Requisition Console) Enables employers and university placement officers to author and publish authenticated internship openings with mandatory schema validation, Fair-Stipend enforcement, and cryptographic posting signatures.
Module 2:	Browse & Search Verified Opportunities (Candidate Catalog Explorer & Audit Console) Enables student job seekers and auditors to explore all active verified opportunities in real time, execute dynamic keyword searches with sanitization guardrails, filter by domain/mode, and inspect cryptographic proof digests.
Responsibilities:	Module 1 handles the <i>producer workflow</i> (input capture, Anti-XSS sanitization, cryptographic block stamping, shared store commit). Module 2 handles the <i>consumer and audit workflow</i> (catalog querying, multi-facet filtering, eligibility verification against student CGPA 3.78, and bookmarking state).

Part 2 — Member 1: Develop Module 1 (Muhammad Umar Afzaal — 23F-3106)

Responsible Member:	Muhammad Umar Afzaal (Roll No: 23F-3106) File: src/pages/PostOpportunityModule.jsx
Primary Function:	Enterprise and university placement requisition portal for drafting, cryptographically signing, and publishing authenticated internship opportunities with live store integration.
Inputs & Actions:	Captures Position Title, Organization, Domain Category (Security, Software, AI, DevOps), Work Mode (Hybrid, Remote, On-site), Monthly Stipend (PKR), Cutoff CGPA, Skill Badges, and Scope Description. Includes 1-click 'Demo Preset Fill' for rapid evaluation.
Input Validation Rules:	Title: Min 4 characters, angle brackets rejected. Fair-Stipend SLA: Must be numeric and ≥ PKR 30,000 to prevent predatory unpaid postings. Academic Boundary: CGPA cutoff bounded between 2.00 and 4.00. Anti-XSS Defense: Min 20 characters; <script> tags strictly forbidden and stripped before memory commit.
JavaScript Functionality:	Dynamic form validation state machine rendering inline field error banners; automated computation of SHA-256 HMAC cryptographic digest (sha256:4a...) and short block ID (0xVER-...); reactive shared state dispatch; form state reset.
Feedback & Data Display:	Feedback: Mint success alert banner with generated audit block ID and full digest; toast notification. Data Display: 'Live Posted Opportunities' feed beside form immediately rendering newly submitted listings with live status chips.

Part 3 — Member 2: Develop Module 2 (Musa Rehan — 23F-3093)

Responsible Member:	Musa Rehan (Roll No: 23F-3093) File: src/pages/BrowseOpportunitiesModule.jsx
Primary Function:	Dynamic catalog explorer and cryptographic verification audit console allowing students and evaluators to browse, search, multi-facet filter, and inspect verified internship listings.
Input / Search Rules:	• Length Guardrail: Search queries capped at 50 characters to prevent client-side ReDoS. • Query Sanitization: Automatically strips injection tokens (< > { } \) and renders saffron security notice. • Empty State Recovery: Renders clear empty-state visual with a 1-click 'Reset all filters' button.
JavaScript Functionality:	• Multi-Criteria Filtering Engine: Real-time matching across role title, company, description, and technical tags, combined with Domain Pills, Work Mode Pills, and Min Stipend SLA dropdown. • Cryptographic Inspection Modal: Interactive modal displaying full SHA-256 digest, audit block ID, and automatic Academic Eligibility Check (compares candidate CGPA 3.78 vs requirement). • Bookmark State Machine: Interactive bookmark toggle with state persistence and toast feedback.
Feedback & Data Display:	• Feedback: Dynamic count banner ('Showing X of Y Verified Positions'), warning notice on sanitized queries. • Data Display: Responsive card grid with company logos, verified chips, stipend SLA, tags, and inspect triggers.

Part 4 — Integrate Both Modules

Integration Strategy:	Both modules reside within the same project codebase (src/pages/). State is unified in src/App.jsx using a centralized opportunities state array and shared utilities in src/data/opportunitiesData.js. Submissions in Module 1 immediately reflect in Module 2 without page refresh.
Navigation Links:	• Navbar: Added dedicated buttons for 'Browse Catalog' (Module 2) and 'Post Opportunity' (Module 1). • Home Page: Added Activity 3 Parallel Module Integration cards with direct launcher buttons. • Dashboards: Linked both modules directly within Student Portal and Placement Officer Console. • Footer: Added quick links to both modules in site navigation.
Integration Checklist:	[✓] Module 1 opens and works correctly [✓] Module 2 opens and works correctly [✓] Both modules accessible via project interface [✓] Local navigation functions cleanly [✓] Unified Senior Editorial design tokens [✓] Existing Activity 1 & 2 features fully intact

Part 5 — Test Both Modules (Testing Results Table)

Test Case / Action	Module	Expected Result	Actual Result	Status
Submit valid requisition via '1-Click Demo Fill'	Module 1 (Umar)	Action succeeds, calculates audit hash, adds to live list	Computed digest 0xVER-4A12; mint alert shown; added to live feed	PASS
Leave required field empty (clear title or set stipend < 30k)	Module 1 (Umar)	System prevents submission and displays localized error	Intercepted submission; displayed red validation alerts	PASS
Search keyword 'Security' & filter by 'Hybrid'	Module 2 (Musa)	Catalog dynamically updates with matching records	Filtered to 3 positions; result count updated reactively	PASS
Cross-module flow: Post job in M1 & search in M2	Integrated	Newly posted job appears in catalog without refresh	New job immediately searchable in Module 2 with proof	PASS

Testing Issues Discovered & Corrections Made

Module 1 issue discovered:	Stipend field originally accepted negative numbers or sub-market zero-stipend entries, and description field did not block unescaped HTML characters.
Module 2 issue discovered:	Typing backslashes or bracket tokens in search disrupted regex filtering, and searches returning zero items left a blank container with no reset mechanism.
Correction made:	1. In Module 1, enforced Fair-Stipend validation ($\geq$ PKR 30,000) and implemented sanitizeInput() stripping script/HTML tags. 2. In Module 2, implemented validateSearchQuery() bounding query length to 50 chars, stripping illegal tokens, and adding an interactive empty-state reset button.

Part 6 — Security-Aware Development

Module	Possible Misuse or Problem	Proposed Response & Frontend Security Defense
Module 1 (Post Opportunity) Lead: Umar Afzaal	Stored XSS & Exploitative Requisitions: Adversary injects malicious <script> into job description to hijack applicant sessions, or posts unpaid listings violating university guidelines.	Sanitization Engine & Fair-Stipend Boundary: Enforced client-side sanitizeInput() stripping script and tag characters; schema validation requiring minimum PKR 30k stipend; automated generation of tamper-evident SHA-256 HMAC posting signatures (0xVER-...).
Module 2 (Browse Catalog) Lead: Musa Rehan	Search Injection & PII Scraping: Adversary executes ReDoS by inputting pathological regex patterns into the search bar, or scrapes recruiter contact information in bulk.	Query Bounds, Regex Sanitization & Data Minimization: Enforced 50-character query length limit; automated stripping of special injection characters (< > { } \); maintained Data Minimization shielding direct contact info until mutual consent unlock.

Submission Requirements & Student Deliverables Guide

Muhammad Umar Afzaal (Roll No: 23F-3106)	Module 1 Lead (Post Verified Internship Opportunity) - Authored src/pages/PostOpportunityModule.jsx and input validation schema. - Submits: Individual/Group PDF Report (SafeHire_Activity_3_Submission_Report.pdf), GitHub repository link, and screenshots of Module 1 (requisition form filled, validation error, success alert with 0xVER-... digest, live posted list).
Musa Rehan (Roll No: 23F-3093)	Module 2 Lead (Browse & Search Verified Opportunities) - Authored src/pages/BrowseOpportunitiesModule.jsx and search query sanitization. - Submits: Individual/Group PDF Report (SafeHire_Activity_3_Submission_Report.pdf), GitHub repository link, and screenshots of Module 2 (search filter active, domain pills, empty-state recovery, and Cryptographic Verification Audit inspection modal).
Joint Project Deliverables	1. Updated project source code (HTML, CSS, JS/React) with both functional modules integrated into existing semester project. 2. Seamless local execution via npm run dev on http://localhost:3000/ without external database dependencies. 3. Completed testing matrix and security-aware development documentation.